

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year: 2024

Programme: BCSIT

Full Marks: 100

Course: Digital Systems

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section “A”

Very Short Answer Questions

Attempt all the questions. [10×2]

1. Which gates are called universal gates? Why?
2. Form, 1's complement form, and 2's complement form.
3. If an S-R Latch with NAND gate has a 0 on the S input and a 1 on the R input and then S input goes to 1, the latch will be in which state?
4. What is an encoder?
5. What do you mean by triggering of flipflop? Also mention different types of triggering.
6. Write two uses of accumulator.
7. A multiplexer has 16 input lines. How many select lines are required.
8. Find the base x if $(211)_x = (152)_8$
9. What is k-map?
10. What is the function of a status register?

Section “B”

Descriptive Answer Questions

Attempt **any five** questions. [5×10]

11. a. If $A=37$ and $B=-18$, then calculate $B-A$ using 2's Complement concept.
b. Write two advantages of Analog and Digital technology each.
12. Obtain the minimal sum of products expression for the following function and implement the same using only 2-input NOR gates.
 $F(A,B,C,D) = \sum_m(0,2,4,6,8,10,15) + \sum_d(3,11,14)$
13. Differentiate between T and D flip-flop along with its truth table and circuit diagram.
14. Explain full adder circuit. Design full subtractor using decoder and or gates.
15. Design a 4-bit odd parity generator combinational circuit using basic gates.
16. Describe the design of 4 bit ALU.

Section “C”

Long Answer Questions

Attempt **any two** questions. [2×15]

17. Explain Seven Segment Decoder and draw the logical diagram for the outputs ‘a’, ‘c’, and ‘f’.
18.
 - a. Explain the types of counters. Design a 4-bit synchronous up/down counter.
 - b. Explain processor unit. Design a 4 bit arithmetic circuit that performs eight different operations.
19. State the algorithm for the state reduction and perform state reduction on the following state table. Also perform state assignment to the reduced table.

Present State	Next State		Output	
	X = 0	X = 1	X=0	X=1
S1	S1	S2	0	0
S2	S3	S4	0	0
S3	S1	S4	0	0
S4	S5	S7	0	1
S5	S1	S7	0	1
S6	S6	S7	0	1
S7	S1	S7	0	1